E-commerce Sales Prediction:

A project by Om Satyawan Pathak

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In [1]:

```
#Import the required modules:
import pandas as pd
import numpy as np
import joblib

from sklearn.model_selection import train_test_split, KFold
from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer

from sklearn.linear_model import LinearRegression
from sklearn.tree import DecisionTreeRegressor
from sklearn.ensemble import AdaBoostRegressor, RandomForestRegressor
from xgboost import XGBRegressor

from sklearn.metrics import r2_score, mean_squared_error, mean_absolute_error

import warnings
warnings.filterwarnings("ignore")
```

In [2]:

```
#Load the dataset:
df = pd.read_csv("ecommerce_sales_dataset.csv")

df.head()
```

Out[2]:

	order_id	order_date	customer_id	product_category	region	quantity	unit_price	discount	payment_method	delivery_days
0	10001	1/1/2022	1102	Beauty	South	7	373.65	0.28	Wallet	10
1	10002	1/2/2022	1435	Clothing	South	7	47.74	0.09	Card	6
2	10003	1/3/2022	1860	Beauty	East	3	311.28	0.31	COD	6
3	10004	1/4/2022	1270	Electronics	West	5	524.47	0.02	Wallet	6
4	10005	1/5/2022	1106	Clothing	West	5	139.87	0.33	Wallet	4

In [3]:

```
#Inspect the dataset:
# Check the number of rows and columns
print("Dataset Shape:", df.shape)

# Check column names and data types
print("\n\nDataset information:")
df.info()

# Check statistical information
print("\n\nDescription of the dataset:")
df.describe()
```

```
Dataset Shape: (5000, 12)


Dataset information:
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 12 columns):
 #   Column            Non-Null Count  Dtype  
---  ------            --------------  -----  
 0   order_id          5000 non-null   int64  
 1   order_date        5000 non-null   object 
 2   customer_id       5000 non-null   int64  
 3   product_category  5000 non-null   object 
 4   region            5000 non-null   object 
 5   quantity          5000 non-null   int64  
 6   unit_price        5000 non-null   float64
 7   discount          5000 non-null   float64
 8   payment_method    5000 non-null   object 
 9   delivery_days     5000 non-null   int64  
 10  customer_rating   5000 non-null   float64
 11  revenue           5000 non-null   float64
dtypes: float64(4), int64(4), object(4)
memory usage: 468.9+ KB
```

Description of the dataset:

Out[3]:

	order_id	customer_id	quantity	unit_price	discount	delivery_days	customer_rating	revenue
count	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000	5000.000000
mean	12500.500000	1505.701200	4.044800	308.418774	0.179984	6.118800	2.973980	1021.955148
std	1443.520003	290.836902	2.020398	169.259369	0.101404	3.153264	1.157722	825.584219
min	10001.000000	1000.000000	1.000000	15.150000	0.000000	1.000000	1.000000	11.210000
25%	11250.750000	1253.000000	2.000000	161.895000	0.090000	3.000000	2.000000	354.527500
50%	12500.500000	1510.000000	4.000000	309.890000	0.180000	6.000000	3.000000	796.650000
75%	13750.250000	1761.000000	6.000000	455.557500	0.270000	9.000000	4.000000	1515.690000
max	15000.000000	1999.000000	7.000000	599.960000	0.350000	11.000000	5.000000	4119.330000

In [4]:

```
# Check missing values
print("No. of missing values:")
print(df.isnull().sum())

# Check duplicate records
print("\n\nDuplicate Rows:", df.duplicated().sum())

print("\n\nCheck unique values in categorical columns:")
print(df.select_dtypes(include="object").nunique())
```

```
No. of missing values:
order_id            0
order_date          0
customer_id         0
product_category    0
region              0
quantity            0
unit_price          0
discount            0
payment_method      0
delivery_days       0
customer_rating     0
revenue             0
dtype: int64


Duplicate Rows: 0
```

```
Check unique values in categorical columns:
order_date          5000
product_category       4
region                 4
payment_method         3
dtype: int64
```

In [5]:

```python
#Data Cleaning:


#Remove duplicate records
df = df.drop_duplicates()

# Convert order_date to datetime
df["order_date"] = pd.to_datetime(
    df["order_date"],
    errors="coerce"
)

# Remove records with missing target or invalid dates
df = df.dropna(subset=["revenue", "order_date"])

# Remove invalid numerical records
df = df[
    (df["quantity"] > 0) &
    (df["unit_price"] >= 0) &
    (df["discount"].between(0, 1)) &
    (df["revenue"] >= 0)
]

# Reset index
df = df.reset_index(drop=True)

print("Shape after cleaning:", df.shape)
print(df.isnull().sum())
```

```
Shape after cleaning: (5000, 12)
order_id            0
order_date          0
customer_id         0
product_category    0
region              0
quantity            0
unit_price          0
discount            0
payment_method      0
delivery_days       0
customer_rating     0
revenue             0
dtype: int64
```

In [6]:

```python
#Feature Engineering:


# Extract useful date features
df["order_year"] = df["order_date"].dt.year
df["order_month"] = df["order_date"].dt.month
df["order_day"] = df["order_date"].dt.day
df["order_dayofweek"] = df["order_date"].dt.dayofweek

# Separate features and target
X = df.drop(columns=[
    "revenue",
    "order_id",
    "customer_id",
    "order_date",
    "delivery_days",
    "customer_rating"
```

```
])

y = df["revenue"]

# Identify numerical and categorical columns
num_cols = X.select_dtypes(include=np.number).columns.tolist()
cat_cols = X.select_dtypes(include="object").columns.tolist()

print("Numerical Columns:", num_cols)
print("Categorical Columns:", cat_cols)
```

```
Numerical Columns: ['quantity', 'unit_price', 'discount', 'order_year', 'order_month', 'o
rder_day', 'order_dayofweek']
Categorical Columns: ['product_category', 'region', 'payment_method']
```

In [ ]:

```

```

In [ ]:

```

```

In [7]:

```
# Machine Learning part:
```

In [8]:

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.2,
    random_state=42
)

print("X_train:", X_train.shape)
print("X_test:", X_test.shape)
print("y_train:", y_train.shape)
print("y_test:", y_test.shape)
```

```
X_train: (4000, 10)
X_test: (1000, 10)
y_train: (4000,)
y_test: (1000,)
```

In [9]:

```
num_cols = X_train.select_dtypes(include=np.number).columns
cat_cols = X_train.select_dtypes(include="object").columns

scaler = StandardScaler()

X_train[num_cols] = scaler.fit_transform(X_train[num_cols])
X_test[num_cols] = scaler.transform(X_test[num_cols])

# One-hot encode categorical columns
X_train = pd.get_dummies(X_train, columns=cat_cols)
X_test = pd.get_dummies(X_test, columns=cat_cols)

# Ensure both datasets have the same columns
X_test = X_test.reindex(columns=X_train.columns, fill_value=0)

print("Training data shape:", X_train.shape)
print("Testing data shape:", X_test.shape)

print("Missing values in X_train:", X_train.isnull().sum().sum())
print("Missing values in X_test:", X_test.isnull().sum().sum())
print("Missing values in y_train:", y_train.isnull().sum())
print("Missing values in y_test:", y_test.isnull().sum())
```

```
Training data shape: (4000, 18)
```

```
Testing data shape: (1000, 18)
Missing values in X_train: 0
Missing values in X_test: 0
Missing values in y_train: 0
Missing values in y_test: 0
```

In [10]:

```
models = {
    "Linear Regression": LinearRegression(),
    "Decision Tree Regressor": DecisionTreeRegressor(random_state=42),
    "Random Forest Regressor": RandomForestRegressor(
        random_state=42,
        n_estimators=100
    ),
    "AdaBoost Regressor": AdaBoostRegressor(random_state=42),
    "XGBoost Regressor": XGBRegressor(
        random_state=42,
        n_estimators=100
    )
}
```

In [11]:

```
from sklearn.model_selection import KFold, cross_val_score
from sklearn.metrics import mean_absolute_error, root_mean_squared_error, r2_score


kf = KFold(n_splits=5, shuffle=True, random_state=42)

results = {}

for name, model in models.items():

    r2_scores = cross_val_score(
        model, X_train, y_train,
        cv=kf, scoring="r2"
    )

    mae_scores = -cross_val_score(
        model, X_train, y_train,
        cv=kf, scoring="neg_mean_absolute_error"
    )

    rmse_scores = -cross_val_score(
        model, X_train, y_train,
        cv=kf, scoring="neg_root_mean_squared_error"
    )

    results[name] = {
        "R2 Score": r2_scores.mean(),
        "MAE": mae_scores.mean(),
        "RMSE": rmse_scores.mean()
    }

    print(name)
    print("Mean R2 Score:", r2_scores.mean())
    print("Mean MAE:", mae_scores.mean())
    print("Mean RMSE:", rmse_scores.mean())
    print("-" * 40)
```

```
Linear Regression
Mean R2 Score: 0.8672811426142172
Mean MAE: 221.4577159180697
Mean RMSE: 300.8223692814246
----------------------------------------
Decision Tree Regressor
Mean R2 Score: 0.9942467183084036
Mean MAE: 42.27419249999999
Mean RMSE: 62.590424512292635
----------------------------------------
```

```
Random Forest Regressor
Mean R2 Score: 0.9981543806546498
Mean MAE: 23.352572375000005
Mean RMSE: 35.296446780513676
----------------------------------------
AdaBoost Regressor
Mean R2 Score: 0.9479893661115085
Mean MAE: 159.0596835010242
Mean RMSE: 188.23228677799133
----------------------------------------
XGBoost Regressor
Mean R2 Score: 0.9984690818171178
Mean MAE: 22.274192639484404
Mean RMSE: 32.27866915062235
----------------------------------------
```

In [12]:

```
results_df = pd.DataFrame(results).T
results_df = results_df.sort_values(by="R2 Score", ascending=False)

print(results_df)

best_model_name = results_df.index[0]
best_model = models[best_model_name]

print("Best Model:", best_model_name)

#Train the best model:
best_model.fit(X_train, y_train)
```

```
                           R2 Score          MAE         RMSE
XGBoost Regressor          0.998469    22.274193    32.278669
Random Forest Regressor    0.998154    23.352572    35.296447
Decision Tree Regressor    0.994247    42.274192    62.590425
AdaBoost Regressor         0.947989   159.059684   188.232287
Linear Regression          0.867281   221.457716   300.822369
Best Model: XGBoost Regressor
```

Out[12]:

▾ XGBRegressor

i ?

```
XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=None, colsample_bynode=None,
             colsample_bytree=None, device=None, early_stopping_rounds=None,
             enable_categorical=False, eval_metric=None, feature_types=None,
             feature_weights=None, gamma=None, grow_policy=None,
             importance_type=None, interaction_constraints=None,
             learning_rate=None, max_bin=None, max_cat_threshold=None,
             max_cat_to_onehot=None, max_delta_step=None, max_depth=None,
             max_leaves=None, min_child_weight=None, missing=nan,
             monotone_constraints=None, multi_strategy=None, n_estimators=10
```

In [13]:

```
#Evaluate the best model on test-data:
y_pred = best_model.predict(X_test)

print("Best Model:", best_model_name)
print("R2 Score:", r2_score(y_test, y_pred))
print("MAE:", mean_absolute_error(y_test, y_pred))
print("RMSE:", root_mean_squared_error(y_test, y_pred))
```

```
Best Model: XGBoost Regressor
R2 Score: 0.9984260630041332
MAE: 22.586621254634853
RMSE: 32.56475327993813
```

In [14]:

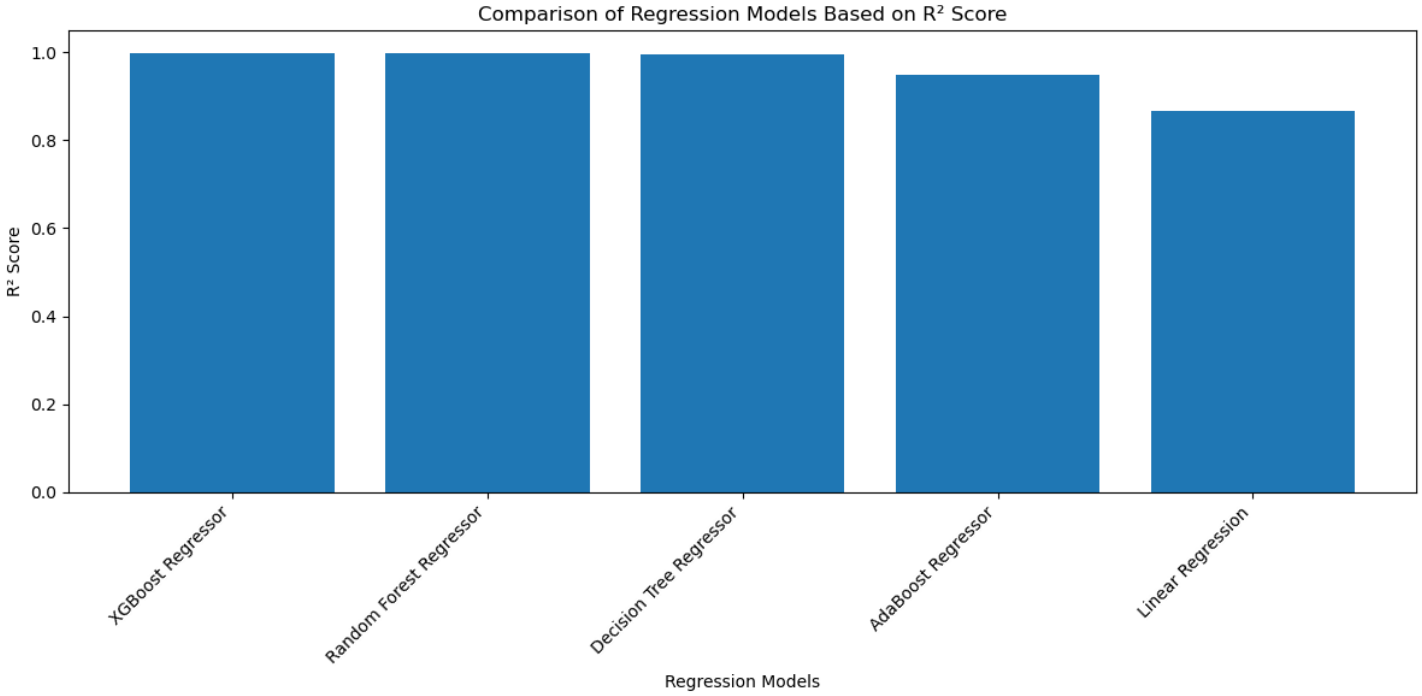
```
#Model Comparison based on R2 Score:
import matplotlib.pyplot as plt

plt.figure(figsize=(12, 6))

plt.bar(results_df.index, results_df["R2 Score"])

plt.title("Comparison of Regression Models Based on R² Score")
plt.xlabel("Regression Models")
plt.ylabel("R² Score")

plt.xticks(rotation=45, ha="right")
plt.tight_layout()
plt.show()
```



In [15]:

```
import mlflow
import mlflow.sklearn

mlflow.set_tracking_uri("http://127.0.0.1:5000")
# Set MLflow experiment name:
mlflow.set_experiment("Complete ECommerce Sales Prediction - All Regression Models Perfor
mance tracking using MLFlow")

# Track each model using the existing models and kf:
for name, model in models.items():

    with mlflow.start_run(run_name=name):

        # Log model name:
        mlflow.log_param("model_name", name)

        # Log model parameters:
        mlflow.log_params({
            key: str(value)
            for key, value in model.get_params().items()
            if isinstance(value, (str, int, float, bool, type(None)))
        })

        # Reuse the existing KFold object:
        r2_scores = cross_val_score(
            model, X_train, y_train,
            cv=kf, scoring="r2"
        )
```

```
        mae_scores = -cross_val_score(
            model, X_train, y_train,
            cv=kf, scoring="neg_mean_absolute_error"
        )

        rmse_scores = -cross_val_score(
            model, X_train, y_train,
            cv=kf, scoring="neg_root_mean_squared_error"
        )

        # Log average performance metrics:
        mlflow.log_metric("CV_R2_Score", r2_scores.mean())
        mlflow.log_metric("CV_MAE", mae_scores.mean())
        mlflow.log_metric("CV_RMSE", rmse_scores.mean())

        print(f"Tracked in MLflow: {name}")
        print("R2 Score:", r2_scores.mean())
        print("MAE:", mae_scores.mean())
        print("RMSE:", rmse_scores.mean())
        print("-" * 40)

print("All models tracked successfully!")
```

```
2026/10/15 06:07:57 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
2026/10/15 06:07:57 INFO mlflow.tracking.fluent: Experiment with name 'Complete ECommerce
Sales Prediction - All Regression Models Performance tracking using MLFlow' does not exis
t. Creating a new experiment.
2026/10/15 06:07:57 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
2026/10/15 06:07:57 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
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2026/10/15 06:07:59 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
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2026/10/15 06:07:59 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
```

```
Tracked in MLflow: Linear Regression
R2 Score: 0.8672811426142172
MAE: 221.4577159180697
RMSE: 300.8223692814246
----------------------------------------
 View run Linear Regression at: http://127.0.0.1:5000/#/experiments/2/runs/1509033edf124
195ba34f7f17e7328d2
 View experiment at: http://127.0.0.1:5000/#/experiments/2
```

```
2026/10/15 06:07:59 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
2026/10/15 06:07:59 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
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2026/10/15 06:07:59 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
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2026/10/15 06:07:59 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
2026/10/15 06:07:59 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
```

2026/10/15 06:08:00 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:00 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:00 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:00 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range

Tracked in MLflow: Decision Tree Regressor
R2 Score: 0.9942467183084036
MAE: 42.27419249999999
RMSE: 62.590424512292635

 View run Decision Tree Regressor at: http://127.0.0.1:5000/#/experiments/2/runs/f7135c1a5559422facb35e7eeafb745d
 View experiment at: http://127.0.0.1:5000/#/experiments/2

2026/10/15 06:08:39 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:40 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:40 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:40 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
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2026/10/15 06:08:40 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:40 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:40 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range

Tracked in MLflow: Random Forest Regressor
R2 Score: 0.9981543806546498
MAE: 23.352572375000005
RMSE: 35.296446780513676

 View run Random Forest Regressor at: http://127.0.0.1:5000/#/experiments/2/runs/bd47c961857546ec98e195761bac7f94
 View experiment at: http://127.0.0.1:5000/#/experiments/2

2026/10/15 06:08:46 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:47 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:47 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:47 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
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2026/10/15 06:08:47 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:47 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range

Tracked in MLflow: AdaBoost Regressor
R2 Score: 0.9479893661115085
MAE: 159.0596835010242
RMSE: 188.23228677799133

 View run AdaBoost Regressor at: http://127.0.0.1:5000/#/experiments/2/runs/19d957b9aad447afb2d85f1c53fafdaf
 View experiment at: http://127.0.0.1:5000/#/experiments/2

2026/10/15 06:08:50 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range
2026/10/15 06:08:50 WARNING mlflow.tracking.request_header.registry: Encountered unexpected error during resolving request headers: list index out of range

```
2026/10/15 06:08:50 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
2026/10/15 06:08:50 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
2026/10/15 06:08:50 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
```

```
Tracked in MLflow: XGBoost Regressor
R2 Score: 0.9984690818171178
MAE: 22.274192639484404
RMSE: 32.27866915062235
----------------------------------------
 View run XGBoost Regressor at: http://127.0.0.1:5000/#/experiments/2/runs/8057c1b005d54
be0abcfc65786c26c7e
 View experiment at: http://127.0.0.1:5000/#/experiments/2
All models tracked successfully!
```

In [16]:

```python
import mlflow
import os

print("Tracking URI:", mlflow.get_tracking_uri())
print("Notebook directory:", os.getcwd())

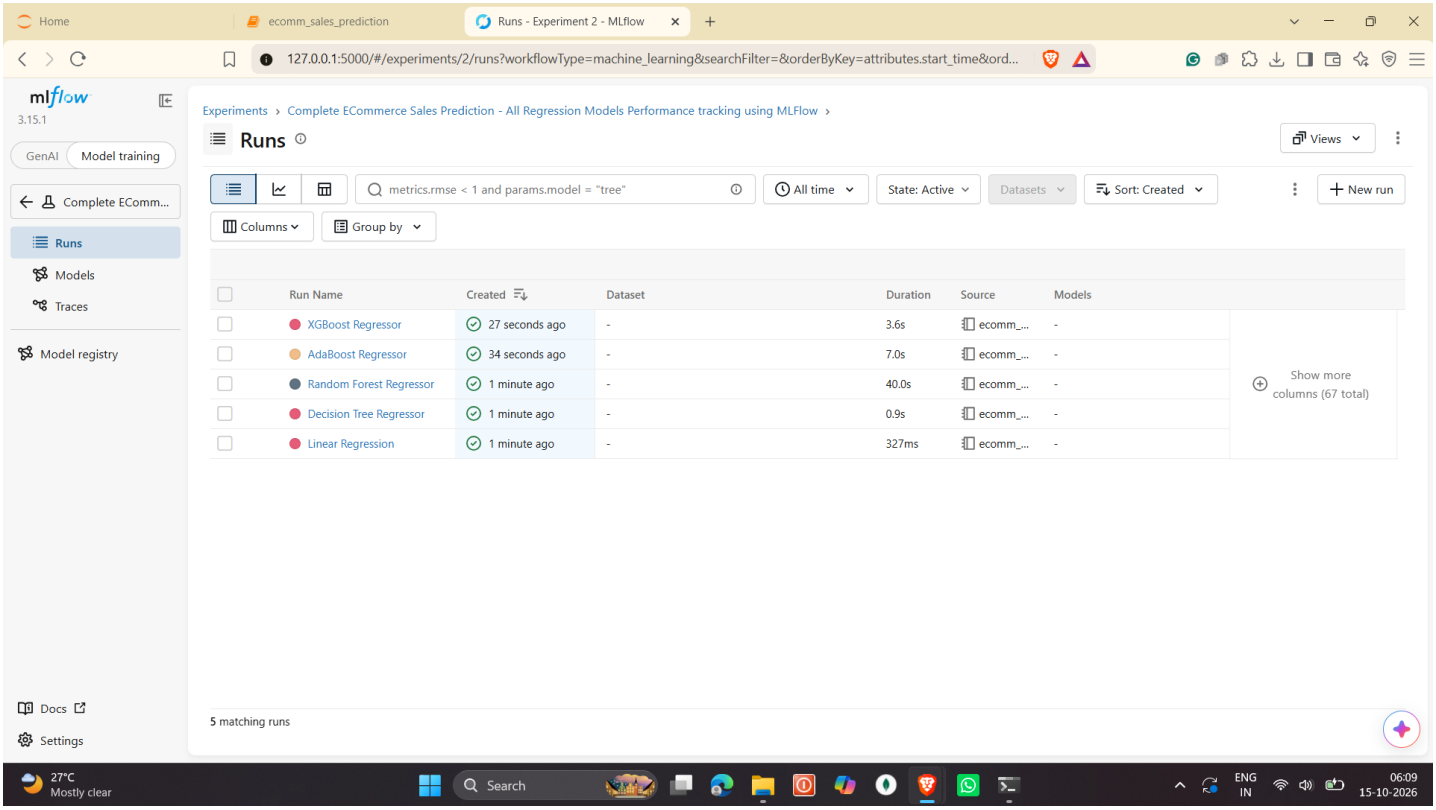
print("\nExperiments visible to the notebook:")

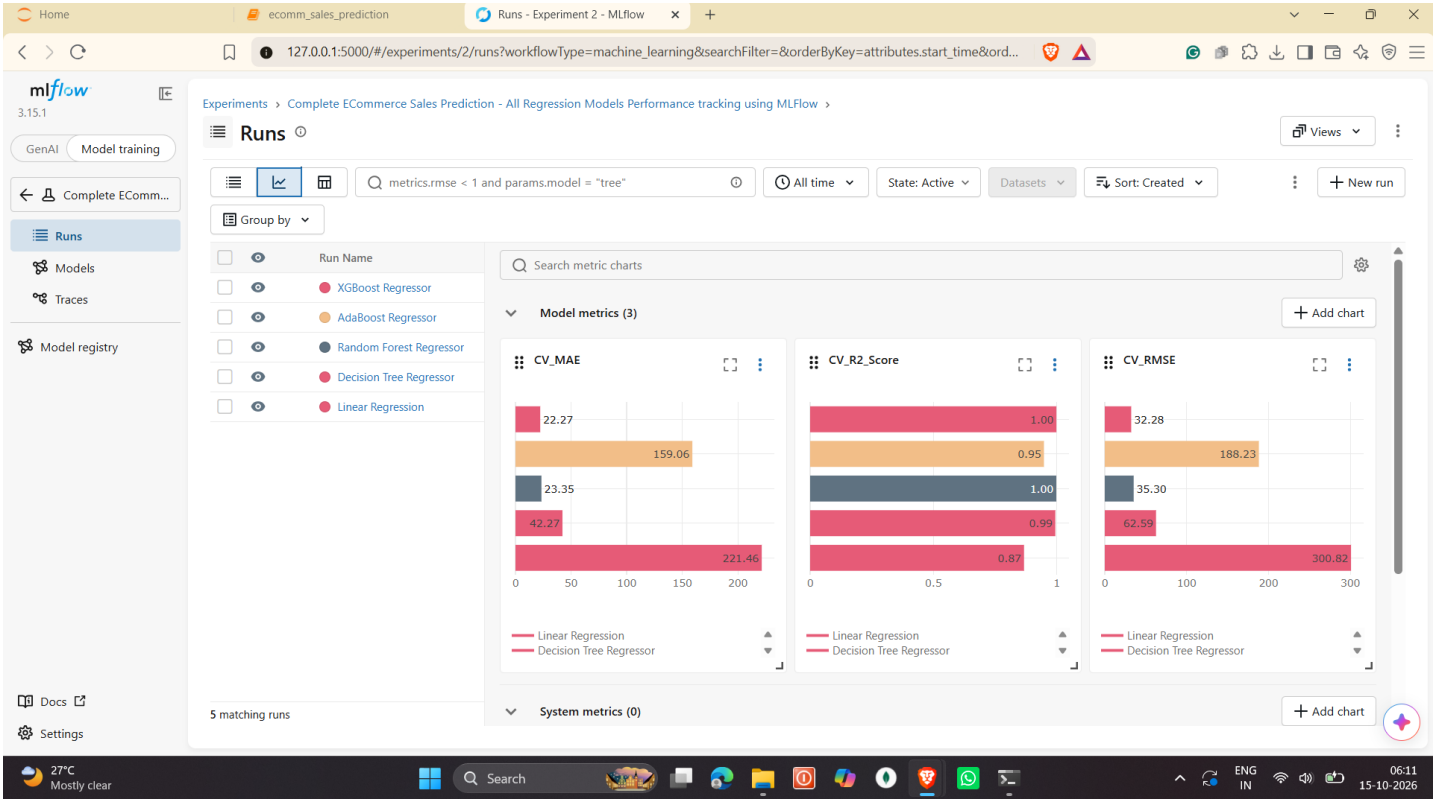
for exp in mlflow.search_experiments():
    print(exp.name, "| ID:", exp.experiment_id)
```

```
2026/10/15 06:09:05 WARNING mlflow.tracking.request_header.registry: Encountered unexpect
ed error during resolving request headers: list index out of range
```

```
Tracking URI: http://127.0.0.1:5000
Notebook directory: C:\Users\Lenovo\Desktop\SELF\PROJECTS ON GITHUB\Data Science PART 2\E
commerce Sales Prediction

Experiments visible to the notebook:
Complete ECommerce Sales Prediction - All Regression Models Performance tracking using ML
Flow | ID: 2
ECommerce Sales Prediction - All Regression Models Performance tracking using MLFlow | ID
: 1
Default | ID: 0
```





In [17]:

```
#Make the best model to do predictions for new data:


new_data = pd.DataFrame({
    "product_category": ["Beauty"],
    "region": ["East"],
    "quantity": [3],
    "unit_price": [311.28],
    "discount": [0.31],
    "payment_method": ["COD"],
    "order_year": [2022],
    "order_month": [3],
    "order_day": [1],
    "order_dayofweek": [2]
})

# Scale numerical columns:
new_data[num_cols] = scaler.transform(new_data[num_cols])

# One-hot encode categorical columns:
new_data = pd.get_dummies(new_data, columns=cat_cols)

# Match the training columns:
new_data = new_data.reindex(columns=X_train.columns, fill_value=0)

# Predict revenue:
predicted_revenue = best_model.predict(new_data)

print("Predicted Revenue:", predicted_revenue[0])
```

```
Predicted Revenue: 643.4581
```

In [ ]:

In [ ]:

In [18]:

```
#Make the best model to do predictions for another new data:
```

```
#Make the best model to do predictions for another new data:


new_data = pd.DataFrame({
    "product_category": ["Clothing"],
    "region": ["West"],
    "quantity": [5],
    "unit_price": [139.87],
    "discount": [0.33],
    "payment_method": ["Wallet"],
    "order_year": [2022],
    "order_month": [5],
    "order_day": [1],
    "order_dayofweek": [7]
})

# Scale numerical columns:
new_data[num_cols] = scaler.transform(new_data[num_cols])

# One-hot encode categorical columns:
new_data = pd.get_dummies(new_data, columns=cat_cols)

# Match the training columns:
new_data = new_data.reindex(columns=X_train.columns, fill_value=0)

# Predict revenue:
predicted_revenue = best_model.predict(new_data)

print("Predicted Revenue:", predicted_revenue[0])
```

```
Predicted Revenue: 488.5463
```

In [ ]:

```

```

In [19]:

```
#Save the best-trained model and preprocessing objects:

joblib.dump(best_model, "best_revenue_prediction_model.pkl")

joblib.dump({
    "num_cols": list(num_cols),
    "cat_cols": list(cat_cols),
    "scaler": scaler,
    "feature_columns": list(X_train.columns)
}, "data_preprocessing.pkl")

print("Model and preprocessing objects saved successfully.")
```

```
Model and preprocessing objects saved successfully.
```

In [ ]:

```

```

In [ ]:

```

```